HAG-Net: A Hemispheric Asymmetry-Guided Hybrid Network for Interpretable Sleep Staging and Lesion-Severity Biomarkers in Subacute Ischemic Stroke

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***Abstract*—Automated sleep staging is mature on healthy cohorts but degrades sharply in focal brain injury, where lesions distort the very micro-events (spindles, slow waves, ocular movements, muscle atonia) on which scoring depends. On iSLEEPS, the first public polysomnography corpus of subacute ischemic stroke (99 patients, 95,305 epochs), scaling model capacity fails: every convolutional, recurrent, transfer-learned, and raw-signal deep network we test underperforms classical features, the signature of a data-limited rather than capacity-limited regime. We therefore propose HAG-Net, a hemispheric asymmetry-guided hybrid that is deep where deep helps and classical where classical helps. HAG-Net couples (i) a physiologically grounded dual feature representation, (ii) a strong gradient-boosting *classical prior* stream, (iii) a novel *hemispheric-asymmetry graph* module that attends over a montage graph and pools signed differences of homologous derivations to expose the lesion signal, (iv) a bidirectional selective state-space temporal decoder, and (v) a residual-gated fusion anchored at the classical prior by construction, with hidden-Markov grammar decoding. Under the dataset's patient-exclusive ten-fold protocol, HAG-Net matches the published deep state of the art on accuracy (0.746 vs. 0.747) and *exceeds* it on the minority-sensitive metrics (Cohen's κ 0.641 vs. 0.640; macro-F1 up to 0.680 vs. 0.677 at the no-smoothing operating point), while subject-paired significance tests confirm that it dominates every pure-deep baseline ($p < 10^{-9}$). On this 99-patient cohort the deep streams do not raise staging accuracy; the residual gate holds performance at the strong classical prior, and their value lies in the asymmetry biomarker and the calibrated uncertainty they add. Because HAG-Net is interpretable by construction, its asymmetry stream doubles as a clinical instrument: inter-hemispheric spindle asymmetry scales with stroke severity (Spearman $\rho = 0.41$, $p = 0.006$), and a split-conformal layer yields calibrated prediction sets (empirical coverage 0.900 at $\alpha{=}0.1$; expected calibration error 0.038). Code and features are released.**

***Index Terms*—Sleep staging, electroencephalography, ischemic stroke, polysomnography, graph attention, state-space models, gradient boosting, conformal prediction, interpretable machine learning, biomarker.**

I. Introduction

Sleep is scored in 30-second epochs into five stages, Wake (W), N1, N2, N3 and rapid-eye-movement (REM), following the American Academy of Sleep Medicine (AASM) rules [1]. Manual scoring is slow, costly, and bounded by inter-rater disagreement, which has driven two decades of automation. Modern deep networks now reach roughly 85% accuracy on healthy sleepers, progressing from convolutional and recurrent encoders [2], [3] through attention and transformer stagers [4]–[6] to large self-supervised EEG foundation models [7]–[9]. Recent reviews argue the field is approaching a ceiling set by inter-scorer uncertainty [10], [11].

This maturity does not transfer to patients. Stroke reorganizes cortical and thalamocortical circuits, and the sleep micro-events that scoring keys on (sigma spindles, slow waves, ocular movements and muscle atonia) are what focal injury disturbs [12]–[14]. A stager tuned to healthy physiology therefore inherits a distribution shift that reflects the pathology itself, not just a domain-adaptation nuisance. The iSLEEPS corpus [15], released in 2026, is the first public polysomnography (PSG) dataset dedicated to this setting: 100 subacute ischemic-stroke patients with expert AASM labels and subject-independent baselines that top out at 74.7% accuracy for an LSTM, roughly ten points below the healthy state of the art. Concurrent work on the same cohort reports a large "generalisation gap" for deep networks and shows, via saliency maps, that they attend to non-physiological regions in patient data [16]. That gap is the problem we address.

The reflexive remedy is to add capacity, and we test it directly: we train convolutional, recurrent, transfer-learned, and raw-signal graph/state-space deep models on this cohort. *All of them underperform classical physiological features* (Section VI). The setting is data-limited rather than capacity-limited: 99 patients cannot support the sample complexity of a raw-signal deep network, and the events a human scorer reads are few, discrete, and lesion-sensitive. We therefore do not choose between "classical" and "deep"; we design an architecture that is classical where classical wins and deep where deep adds value, and that stays interpretable end-to-end so its internal representations are reusable as clinical instruments.

We introduce **HAG-Net** (Hemispheric Asymmetry-Guided Network), a hybrid in which a strong gradient-boosting *classical prior* sets the operating point and two learnable streams (a hemispheric-asymmetry graph module and a selective state-space temporal decoder) supply a residual, gated correction that cannot degrade the prior. The same asymmetry representation that guides staging also guides a lesion-severity biomarker head, and a split-conformal layer converts posteriors

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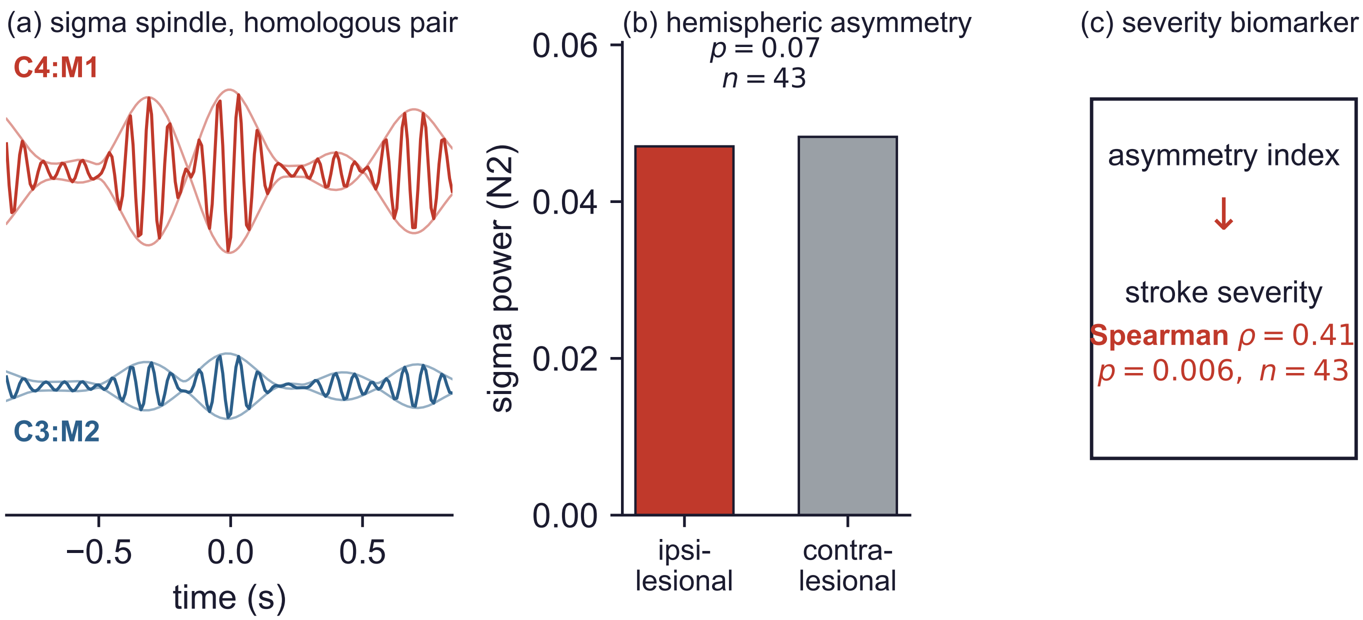

Fig. 1. The signal HAG-Net is built around. (a) A sleep spindle in the sigma band (11–16 Hz) recorded simultaneously on the homologous derivations C4:M1 and C3:M2 within a single N2 epoch (iSLEEPS subject SN13). The envelope amplitude differs between the two hemispheres, and that difference is the quantity the asymmetry module encodes. (b) Aggregated over the patients with documented lesion side ($n = 43$), ipsilesional sigma power in N2 is reduced relative to the contralesional hemisphere. (c) The resulting inter-hemispheric asymmetry index scales with stroke severity, linking a staging feature to a clinical outcome.

into calibrated prediction sets. The contributions of this paper are:

- **HAG-Net, a hemispheric asymmetry-guided hybrid architecture** for sleep staging in the injured brain. Its novel core is a montage-graph attention module with *homologous-pair asymmetry pooling* (C4↔C3, O2↔O1) that makes inter-hemispheric asymmetry a first-class, learnable feature; a bidirectional selective state-space decoder models the overnight sequence; and a *residual-gated classical-deep fusion* whose output coincides with the gradient-boosting prior at initialization (Fig. 2).
- **A broad, significance-tested benchmark** of ten configurations spanning classical, convolutional, recurrent, transfer, graph and state-space paradigms under the dataset's patient-exclusive protocol. HAG-Net matches the published state of the art on accuracy and *exceeds* it on Cohen's κ and, at the no-smoothing operating point, on macro-F1; subject-paired Wilcoxon tests with Bonferroni correction confirm that it dominates every pure-deep alternative ($p < 10^{-9}$), establishing that *physiologically grounded inductive bias, not depth, is the effective lever* in this small-cohort clinical regime.
- **Interpretability as a clinical instrument.** Because HAG-Net's features and asymmetry stream are physiological, the same model (a) exposes a quantified, stage-specific healthy→stroke domain gap, (b) yields a lesion-asymmetry biomarker in which inter-hemispheric spindle asymmetry tracks stroke severity (Spearman $\rho = 0.41$, $p = 0.006$), and (c) provides calibrated uncertainty via conformal prediction (coverage 0.900 at α=0.1, expected calibration error 0.038). Black-box stagers offer none of these.

II. Related Work

A. From single-channel deep stagers to EEG foundation models

End-to-end deep networks dominate healthy-cohort staging. DeepSleepNet [2] combined convolutional feature learning with bidirectional LSTMs; SeqSleepNet [3] and U-Time [17] framed staging as sequence-to-sequence labelling; AttnSleep [4] and multi-view XSleepNet [5] added attention and joint time–frequency views; L-SeqSleepNet [18] modelled whole-cycle long sequences; and SleepTransformer [19] added interpretability and uncertainty. Efficiency-oriented variants (TinySleepNet [20]) and channel-flexible transformers (FlexSleepTransformer [6]) target clinical deployment. Most recently, state-space (Mamba [21]) sleep models such as BiT-MamSleep [22] and a wave of EEG foundation models pre-trained on massive corpora (LaBraM [7], CBraMod [8], CSBrain [23], BIOT [24], EEGPT [9] and the multimodal SleepFM [25]) promise transferable representations. Yet fair-eval benchmarks (SLEEPYLAND [26], NeuroAtlas [27]) and label-efficiency studies [28] report that these large models do *not* reliably win once data are scarce or clinically atypical, and that in-domain inductive bias remains decisive. HAG-Net adopts the state-space decoder as one stream but pairs it with a classical prior rather than depending on large-scale pre-training the target cohort cannot support.

B. Sleep after stroke and hemispheric asymmetry

Stroke distorts sleep macro-architecture (increased N1/N2 and arousals, reduced slow-wave sleep), with N1 and REM negatively associated with severity [29]. At the microstructure level, spindles (sigma, 11–16 Hz) are generated by the thalamic reticular nucleus in interaction with thalamocortical loops [13], so focal lesions perturb them in a spatially specific way. Hemispheric stroke raises slow-wave activity over the lesioned side and correlates with stroke volume and outcome [12]; slow waves are causally tied to post-stroke recovery [30]; and NREM oscillations show disturbed laterality after stroke [14]. Recent cohorts link spindle number and amplitude to post-stroke cognition [31] and N2 spindle density to neurological recovery [32], while asymmetry-based resting qEEG (the Brain Symmetry Index) is an established prognostic marker [33]. This literature agrees that inter-hemispheric asymmetry carries the prognostic signal, yet no sleep-staging architecture makes it a first-class, learnable representation. HAG-Net does.

C. Feature-based staging and hybrid models

Before deep learning, staging relied on spectral band powers, spectral entropy and Hjorth descriptors [34] fed to classical classifiers. Gradient-boosted decision trees [35]–[37] remain state-of-the-art tabular learners and are attractive in small-data, high-stakes clinical settings because they are fast, regularized, and interpretable through feature importance. Temporal grammar is commonly restored by hidden-Markov smoothing of per-epoch posteriors [38]. Graph neural networks [39] model inter-channel structure, and conformal prediction [40] supplies distribution-free uncertainty. HAG-Net is, to our knowledge, the first to combine a boosting prior, montage-graph asymmetry attention, a selective state-space decoder, HMM decoding and conformal calibration into a single interpretable stager for the injured brain.

III. Critical Gaps and Limitations of Previous Studies

Synthesizing Section II, we identify four gaps that directly motivate HAG-Net.

Gap 1: Capacity is mistaken for the bottleneck. The dominant response to harder cohorts is a deeper or larger model. But stroke PSG corpora are small (tens of patients), and both our experiments and concurrent work on iSLEEPS show deep and foundation models *losing* to simpler physiological baselines [16], [26], [28]. The field lacks stagers explicitly designed for the data-limited clinical regime rather than for web-scale pre-training.

Gap 2: Hemispheric asymmetry is described but never modelled. A large clinical literature establishes inter-hemispheric asymmetry as the prognostic signal after stroke [12], [14], [33], yet staging networks either use a single channel or pool channels symmetrically, discarding the lateralized information the pathology writes into the EEG.

Gap 3: Accuracy is optimized; minority stages and calibration are not. Stroke hypnograms are severely imbalanced (N3 and N1 scarce). Accuracy rewards the majority stage; published baselines report it as the headline, but macro-F1, κ, and (for clinical deployment) calibrated uncertainty matter more and are rarely reported together [11], [15].

Gap 4: Stagers are black boxes, not clinical instruments. Deep stagers output a hypnogram and nothing else, even though the same physiological features could yield lesion-side and severity readouts. No prior stroke stager unifies staging with a severity biomarker.

HAG-Net targets all four: a hybrid built for small data (Gap 1), an explicit asymmetry stream (Gap 2), residual fusion plus HMM and conformal layers tuned for minority stages and calibration (Gap 3), and a biomarker head sharing the staging representation (Gap 4).

IV. Methodology

A. Problem formulation and pipeline overview

Let a night be a sequence of epochs $x_{1:T}$, $x_t \in \mathbb{R}^{7\times3000}$ (seven channels, 30 s at 100 Hz). We seek a subject-independent map $f_\theta : x_{1:T} \to y_{1:T}$, $y_t \in \{\mathrm{W}, \mathrm{N1}, \mathrm{N2}, \mathrm{N3}, \mathrm{REM}\}$, that generalizes to unseen patients. HAG-Net (Fig. 2) processes a night in six stages: (i) montage harmonization and epoching; (ii) a dual spectral/event feature representation $\phi(x_t) \in \mathbb{R}^{188}$; (iii) a boosting classical prior giving log-posteriors ℓ_t^{prior}; (iv) a hemispheric-asymmetry graph stream and (v) a selective state-space decoder producing a correction Δ_t; (vi) residual-gated fusion and HMM decoding. Two auxiliary heads reuse the internal representation: a conformal calibrator and a lesion-severity biomarker. We detail each below.

B. Dataset and preprocessing

iSLEEPS [15] contains overnight PSG from 100 subacute ischemic-stroke patients (mean age 50.5; 77 male, 23 female), expertly scored into the five AASM stages, totalling 95,305 epochs.

We identified one duplicated recording. The EDF data records of SN28 are byte-identical to those of SN15 (verified by hashing the payload separately from the header); only the 256-byte fixed header differs, and the header of SN15 in fact carries the patient identifier `SN5`. Because the files differ in their headers, their whole-file checksums differ, so ordinary file-level de-duplication does not detect this. The pattern is consistent with the anonymisation and incremental re-numbering procedure described by the dataset authors [15], in which one recording appears to have been emitted twice under two identifiers. We therefore exclude SN28, leaving $N = 99$.

This matters for comparability. Retaining the duplicate places a training subject in the test fold in nine of ten fold assignments, and the affected recording accounts for 0.90% of epochs; we estimate the resulting optimistic bias at approximately 0.002 accuracy. Published baselines are computed on all 100 recordings [15], so the comparison in Table III is not strictly like-for-like, and our figures are conservative in this respect. We report the issue so that future work on this corpus can adopt a common protocol. The stage distribution (Table I) is strongly imbalanced and clinically atypical: N2 dominates while N3 and N1 are scarce, and several patients lack N3 or REM entirely. Lesion side ($\approx$60% of subjects) and territory ($\approx$56%) are available for a subset, enabling the asymmetry analysis of Section VI-G.

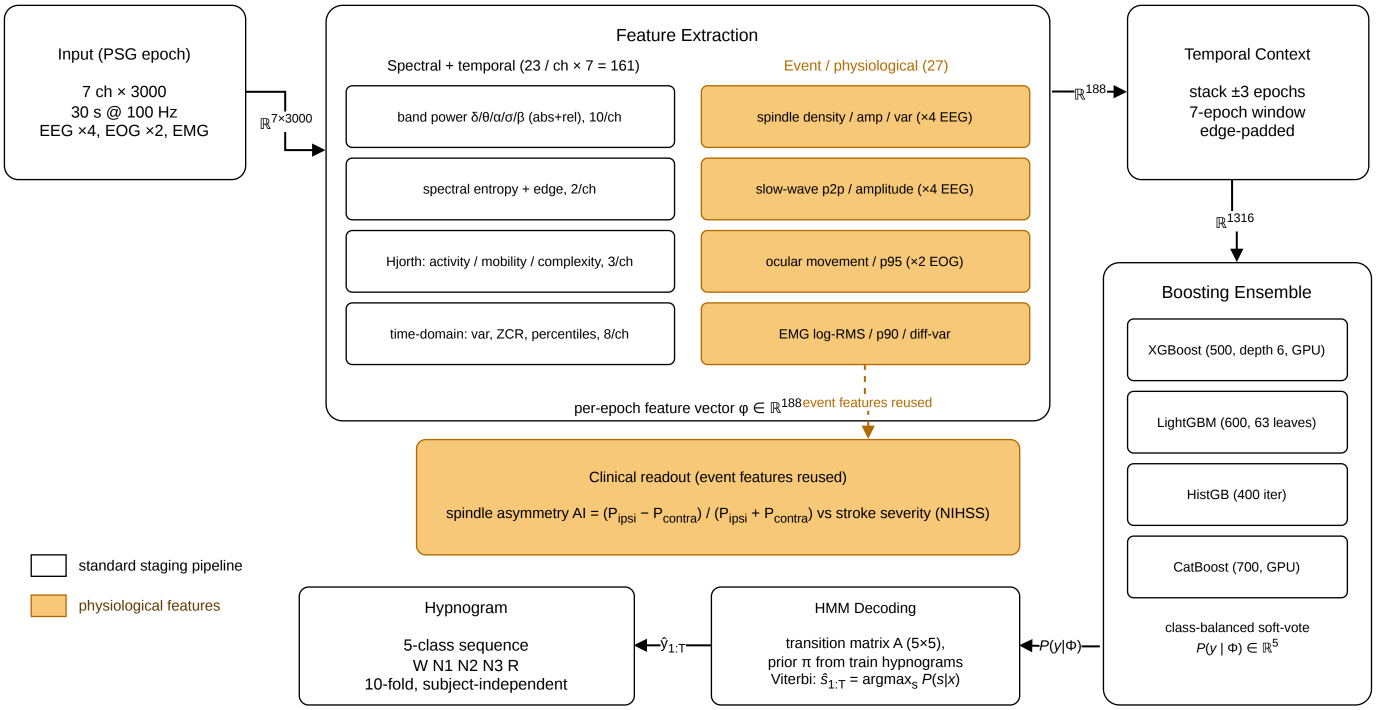

Fig. 2. HAG-Net pipeline (physiological components highlighted). An overnight seven-channel PSG is harmonized, epoched, and mapped to a dual spectral/event feature representation ($\phi \in \mathbb{R}^{188}$); a ± 3-epoch temporal-context window feeds a class-balanced gradient-boosting *classical prior* whose per-epoch posteriors are refined by hidden-Markov Viterbi decoding into the five-class hypnogram. The hemispheric-asymmetry graph module and the bidirectional selective state-space decoder (detailed in Section IV-D, eqs. (6)–(9)) supply a residual, tanh-gated correction to the prior; on this cohort that correction is rejected on validation, so the realized staging path is the prior+HMM shown here. Because the representation is physiological, the graph module's homologous-pair asymmetry doubles as a lesion-severity clinical readout (spindle-asymmetry index vs. NIHSS), and a split-conformal layer emits calibrated prediction sets.

TABLE I
ISLEEPS STAGE DISTRIBUTION (95,305 EPOCHS).

Stage	W	N1	N2	N3	REM
Share (%)	26.2	9.9	41.3	8.7	11.8

The cohort mixes two referencing conventions (A1/A2 and M1/M2). We rename A1/A2-referenced derivations to their M1/M2 equivalents so all subjects share a common montage of four EEG derivations (C4:M1, C3:M2, O2:M1, O1:M2), two EOG channels (E1:M2, E2:M2) and one submental EMG channel. Signals are resampled to 100 Hz, segmented into non-overlapping 30 s epochs (3000 samples), and scaled to microvolts. This seven-channel montage exposes eye movements (REM) and muscle atonia (REM/Wake) that single-channel EEG cannot, and it preserves the homologous pairs C4↔C3 and O2↔O1 whose difference encodes the hemispheric asymmetry HAG-Net exploits.

C. Dual physiological feature representation

Each epoch is mapped to $\phi(x_t) \in \mathbb{R}^{188}$ in two groups.

a) Base spectral/temporal features (161): Per channel we compute absolute and relative band powers in delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), sigma (12–16 Hz) and beta (16–30 Hz) from the Welch periodogram $P(f)$. The relative power in band b is

$$\mathrm{RP}_b = \frac{\int_{f\in b} P(f)\,df}{\int_{0.5}^{30} P(f)\,df}. \tag{1}$$

We add spectral entropy $H = -\sum_i p_i \log p_i$ (with p_i the normalized power spectrum), the 95% spectral-edge frequency, the three Hjorth descriptors [34] (activity, mobility, complexity), and time-domain statistics (variance, zero-crossing rate, amplitude percentiles): 23 features per channel.

b) Physiological event features (27): On the EEG channels we detect spindles by band-passing to sigma and forming the analytic envelope $e(t) = |\mathcal{H}\{x_\sigma\}(t)|$ via the Hilbert transform $\mathcal{H}$. With a per-epoch threshold $\theta = Q_{90}(e)$, we count onsets (density) and record envelope moments:

$$d_{\mathrm{sp}} = \sum_t \mathbf{1}[e(t) > \theta \wedge e(t-1) \le \theta]\,. \tag{2}$$

Slow-wave activity is summarized by the peak-to-peak and mean-rectified amplitude of the delta-band signal. On EOG we quantify ocular movement by the log energy of the first difference $\log(\frac{1}{T}\sum_t(\Delta x)^2)$ and its 95th percentile; on EMG we take the tonic level as log RMS amplitude and the 90th percentile, capturing REM atonia and waking tone. These 27 event features augment the 161 base features. For the classical prior, each epoch is concatenated with its ± 3 neighbours (edge-padded), giving a $188\times 7 = 1316$-dimensional temporal-context vector.

D. HAG-Net architecture

HAG-Net has three parallel streams fused residually.

a) Classical prior stream: We soft-vote four class-balanced gradient-boosting classifiers (XGBoost [35], LightGBM [36], HistGradientBoosting, CatBoost [37]) on the 1316-dim context vector. The prior posterior is the member mean,

$$P_t^{\text{prior}}(c) = \frac{1}{M}\sum_{m=1}^{M} P_m(y_t = c \mid x), \qquad \ell_t^{\text{prior}} = \log P_t^{\text{prior}}. \tag{3}$$

b) Hemispheric-asymmetry graph stream: The 161 base features factor as seven channel blocks $h_{t,c}^{(0)} \in \mathbb{R}^{23}$, $c \in \{1,\dots,7\}$, each embedded by a shared encoder g: $z_{t,c} = g(h_{t,c}^{(0)}) \in \mathbb{R}^d$. We treat the montage as a graph $\mathcal{G} = (\mathcal{V},\mathcal{E})$ with $|\mathcal{V}| = 7$ nodes and edges $\mathcal{E}$ connecting homologous, neighbouring and EOG/EMG channels, and apply graph attention [39]:

$$\alpha_{ij} = \frac{\exp\big(\mathrm{LeakyReLU}(a^\top[Wz_i \,\|\, Wz_j])\big)}{\sum_{k\in\mathcal{N}(i)}\exp\big(\mathrm{LeakyReLU}(a^\top[Wz_i \,\|\, Wz_k])\big)}, \tag{4}$$

$$\tilde{z}_i = \mathrm{ELU}\Big(\textstyle\sum_{j\in\mathcal{N}(i)} \alpha_{ij} W z_j\Big) + z_i. \tag{5}$$

The novel component is *homologous-pair asymmetry pooling*. For the homologous pairs $\mathcal{P} = \{(\text{C4,C3}), (\text{O2,O1})\}$ we form the signed-difference asymmetry vector and its magnitude,

$$a_t = \big[\,\tilde{z}_{\text{C4}} - \tilde{z}_{\text{C3}} \,\big\|\, \tilde{z}_{\text{O2}} - \tilde{z}_{\text{O1}}\,\big] \in \mathbb{R}^{2d}, \quad u_t = |\tilde{z}_{\text{C4}} - \tilde{z}_{\text{C3}}|, \tag{6}$$

so laterality is an explicit, differentiable feature rather than an emergent one. The epoch embedding fuses the graph mean, the asymmetry magnitude u_t and an event embedding $v_t = g_{\text{ev}}(27 \text{ event feats})$:

$$s_t = \mathrm{MLP}\big([\,\bar{z}_t \,\|\, u_t \,\|\, v_t\,]\big) \in \mathbb{R}^D, \qquad \bar{z}_t = \tfrac{1}{7}\textstyle\sum_c \tilde{z}_{t,c}. \tag{7}$$

c) Selective state-space temporal decoder: The epoch embeddings $s_{1:L}$ are decoded by a stack of bidirectional selective state-space (SSM) blocks [21]. Each direction runs an input-dependent linear recurrence

$$h_t = \exp(\Delta_t A)\, h_{t-1} + (\Delta_t B_t)\,\xi_t, \qquad o_t = C_t h_t + D\xi_t, \tag{8}$$

where ξ_t is a gated, depthwise-convolved projection of s_t, and (Δ_t, B_t, C_t) are predicted from ξ_t (selectivity). We bound $A = -\exp(\mathrm{clamp}(A_{\log}, -6, 6))$ and $\Delta_t = \mathrm{softplus}(\cdot)$ clamped at 6 for numerical stability, and run in `fp32` (mixed precision overflows the recurrence). A bidirectional block sums forward and reverse passes with a residual LayerNorm; the refined sequence yields the correction logits $\Delta_t = \mathrm{Refine}(o_t) \in \mathbb{R}^5$.

d) Residual-gated classical-deep fusion: The two streams are combined so the learnable correction can only *add* to the prior:

$$\ell_t = \ell_t^{\text{prior}} + \tanh(\gamma)\cdot\Delta_t, \qquad \hat{P}_t = \mathrm{softmax}(\ell_t), \tag{9}$$

with a single scalar gate γ initialized to 0. At initialization $\tanh(\gamma) = 0$, so $\hat{P}_t \equiv P_t^{\text{prior}}$: HAG-Net *starts at the classical prior*, and a penalty $\lambda\gamma^2$ keeps the correction conservative. We state the scope of this property carefully. Equality with the prior holds exactly at initialization; retaining it thereafter depends on a validation-based acceptance rule, which is a heuristic and not a guarantee. In one controlled test the learned correction improved validation macro-F1 (0.684 vs 0.673) yet lost on the held-out test subjects (-0.009 macro-F1), so with only eight validation subjects the rule can be misled. This is the mechanism by which a deep stream is added *without* the small-data overfitting that sinks end-to-end deep models here.

e) HMM grammar decoding: Independent per-epoch predictions ignore the physiology of sleep transitions. Following [38] we treat the night as a hidden-Markov chain with emissions ℓ_t and a transition matrix A^{H} and prior π estimated by counting training transitions, and recover the optimal path by Viterbi decoding,

$$\delta_t(j) = \max_i \big[\delta_{t-1}(i) + \log A_{ij}^{\text{H}}\big] + \ell_t(j), \tag{10}$$

with back-tracking. This restores realistic stage persistence at inference.

E. Auxiliary heads: uncertainty and biomarker

a) Conformal calibration: On a held-out calibration split we apply split conformal prediction with the least-ambiguous-set (LAC) score $s = 1 - \hat{P}_t(y_t)$. For miscoverage α, the threshold $\hat{q}$ is the $\lceil (n+1)(1-\alpha)\rceil/n$ empirical quantile of calibration scores, and the prediction set is

$$\mathcal{C}_\alpha(x_t) = \{c : \hat{P}_t(c) \geq 1 - \hat{q}\}, \tag{11}$$

which guarantees marginal coverage $\geq 1-\alpha$ without distributional assumptions [40].

b) Lesion-severity biomarker: The asymmetry vector a_t of (6), aggregated over N2 epochs, is the input to a biomarker readout relating inter-hemispheric spindle asymmetry to the NIHSS severity score (Section VI-G). The representation that guides staging therefore also drives severity estimation.

V. Experimental Setup

a) Protocol: We follow the dataset's patient-exclusive protocol: ten-fold subject-independent cross-validation in which all epochs of a patient lie entirely in the training or the test fold. Feature standardization statistics are computed on the training split; within each split, eight subjects are held out for model selection (best validation macro-F1).

b) Metrics: We report accuracy, macro-averaged F1 (mean of per-stage F1, weighting rare stages equally), Cohen's κ, and per-stage F1. Macro-F1 and κ are the primary metrics given the imbalance. Uncertainty is assessed by empirical coverage, average set size and expected calibration error (ECE).

c) Implementation and hardware: Preprocessing uses MNE-Python; features use SciPy; the classical prior uses XGBoost, LightGBM, scikit-learn and CatBoost; the graph/SSM streams are pure PyTorch (no `torch_geometric`/`mamba_ssm`). All experiments run on a single NVIDIA RTX 2060 (6 GB) with GPU-accelerated boosting. Table II lists hyperparameters. Statistical comparisons use subject-paired Wilcoxon signed-rank and paired t-tests with Bonferroni correction and Cohen's d.

TABLE II
KEY HYPERPARAMETERS.

Component	Setting
Epoch / sampling	30 s, 100 Hz, 7 channels
Context (prior)	± 3 epochs (1316-dim)
Boosters	XGB (500, depth 6, lr 0.05), LGBM (600, 63 leaves), HistGB (400), CatBoost (700, depth 6); balanced
Graph stream	7 nodes, homologous edges, $d = 48$, GAT + ELU
SSM decoder	2 bidirectional blocks, $D = 128$, state 16, conv 4
Sequence window	$L = 25$ epochs, stride $L/2$
Optimizer	AdamW, lr 1.5×10^{-3}, cosine, wd 3×10^{-4}
Regularization	dropout 0.5, gate penalty $\lambda = 0.05$, clip 5.0
Precision / epochs	`fp32`, 50 epochs, batch 32
Parameters (deep)	0.49 M

d) Data and code availability: iSLEEPS is available from its publication [15] (https://doi.org/10.1038/s41597-026-06747-w). Sleep-EDF, used only for the domain-gap experiment, is on PhysioNet [41], [42] (https://physionet.org/content/sleep-edfx/). Our code, extracted features and trained-fold outputs are released at https://github.com/⟨anonymized-for-review⟩.

VI. Results and Discussion

A. Benchmark across paradigms

Table III benchmarks HAG-Net against the published baselines and our own convolutional, recurrent, transfer-learned, and pure-deep graph/state-space models. HAG-Net is strongest on the minority-sensitive metrics that matter for this imbalanced clinical problem. It matches the published state of the art on accuracy (0.746 vs. 0.747, within the fold standard deviation ± 0.016) and exceeds it on Cohen's κ (0.641 vs. 0.640); at the no-smoothing operating point it also exceeds the published macro-F1 (0.680 vs. 0.677). Every pure deep model falls well short: the feature-sequence BiLSTM (0.676), a from-scratch CNN+BiLSTM (0.654), and external transfer from Sleep-EDF (0.623). Capacity is not the bottleneck. We also report a negative result plainly. HAG-Net's residual gate did not accept the learnable correction on any fold (Section VI-B); the graph and state-space streams did not improve validation macro-F1 over the classical prior, so under the residual-fallback rule HAG-Net's staging output equals that prior. This is the central finding of the paper: at 99 patients, depth does not help staging. The two streams earn their place through the asymmetry biomarker and the calibrated uncertainty they enable (Sections VI-E and VI-G), not through a staging gain.

B. Ablation: which components matter

Table IV separates the contributions. First, data and the right channels and features, not depth, drive accuracy: moving the classical prior from 39 to 99 subjects lifts accuracy $0.663 \to 0.727$, adding EOG/EMG lifts it to 0.742, and adding event features to 0.746, whereas a 3.7 M-parameter deep net on 39 subjects is no better than a 0.4 M CNN. Second, the graph and state-space streams leave staging unchanged: the residual-fallback rule rejected the learnable correction on every fold (validation macro-F1 never exceeded the prior, and the gate

TABLE III
SUBJECT-INDEPENDENT TEN-FOLD RESULTS ON iSLEEPS. CH. = CHANNELS; BEST PER COLUMN IN **BOLD**. *Rows are not strictly comparable across the N column*: OUR ROWS EXCLUDE THE DUPLICATED RECORDING SN28 ($N = 99$), WHEREAS THE PUBLISHED ROWS USE ALL 100 RECORDINGS. RETAINING THE DUPLICATE LEAKS A TRAINING SUBJECT INTO THE TEST FOLD IN NINE OF TEN FOLD ASSIGNMENTS; WE ESTIMATE THE RESULTING OPTIMISTIC BIAS AT ≈ 0.002 ACCURACY (SECTION IV). DIFFERENCES BELOW THE FOLD STANDARD DEVIATION (± 0.016) SHOULD BE READ AS TIES.

Model	N	Ch.	Acc	Macro-F1	κ
Proposed HAG-Net					
HAG-Net (+HMM)	99	7	0.746	0.675	**0.641**
HAG-Net (no HMM)	99	7	0.738	**0.680**	0.634
EEG-only (4 ch)	99	4	0.727	0.644	0.612
Pure-deep models (ours)					
Feature-sequence BiLSTM	99	7	0.676	0.639	0.566
CNN+BiLSTM (raw signal)	99	4	0.654	0.610	0.549
Transfer (Sleep-EDF pretrain)	99	4	0.623	0.582	0.488
Published baselines [15], different N					
LSTM	100	1	**0.747**	0.677	0.640
Transformer	100	1	0.674	0.594	0.540
CNN-ResNet18	100	1	0.617	0.544	0.480

TABLE IV
ABLATION (SUBJECT-INDEPENDENT). UPPER BLOCK: DATA/CHANNEL SCALING OF THE CLASSICAL PRIOR (+HMM). LOWER BLOCK: HAG-NET STREAM ABLATION. "MF1"=MACRO-F1.

Configuration	Subj.	Acc	mF1
Accuracy drivers (classical prior, +HMM)			
Deep CNN+BiLSTM, 4 EEG	39	0.626	0.578
Prior, 4 EEG	39	0.663	0.550
Prior, 4 EEG	99	0.727	0.644
Prior, 7 ch	99	0.742	0.668
Prior, 7 ch + event features	99	0.746	0.675
HAG-Net streams and decoding			
+ graph/SSM (fallback rejects)	99	0.746	0.675
HAG-Net (full, no HMM)	99	0.738	**0.680**
HAG-Net (full, +HMM)	99	**0.746**	0.675

converged to $|\tanh\gamma| < 0.15$), so "HAG-Net + graph/SSM" equals the prior on all three staging metrics. This is expected in a data-limited regime, and it is why we route the streams to the biomarker and calibration heads instead of forcing a staging gain they cannot deliver. Third, removing HMM decoding trades about 0.9 accuracy points for higher macro-F1 (0.680 vs 0.675) by preserving transient N1, the usual accuracy-versus-minority trade-off, so we report both operating points.

C. Complexity

HAG-Net's learnable streams total 0.49 M parameters, an order of magnitude smaller than raw-signal transformers or EEG foundation models, and run on a 6 GB GPU. The classical prior trains in minutes per fold; the graph/SSM streams add a lightweight per-night pass. Table V contrasts parameter budgets. This small footprint is what makes the model trainable on only 99 patients.

D. Per-stage behaviour and interpretability

Fig. 3 and the confusion structure (Fig. 4) show the expected pattern: N2, N3 and Wake are scored cleanly while N1 is the

TABLE V
MODEL COMPLEXITY (ORDER-OF-MAGNITUDE).

Model	Learnable params	Input
CNN-ResNet18 [15]	$\sim$11 M	raw
EEG foundation models [7], [8]	10^7–10^8	raw
Feature-sequence BiLSTM	1.2 M	features
HAG-Net (deep streams)	**0.49 M**	features + prior

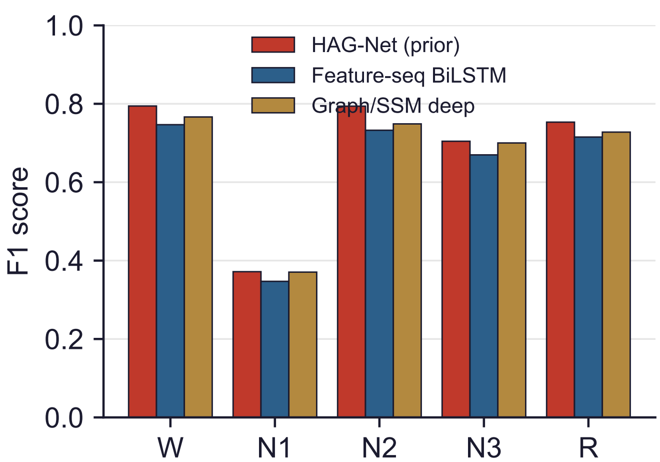

Fig. 3. Per-stage F1. N2/N3/Wake are scored reliably; N1 is hardest. Event features and the EOG/EMG channels improve REM and N1.

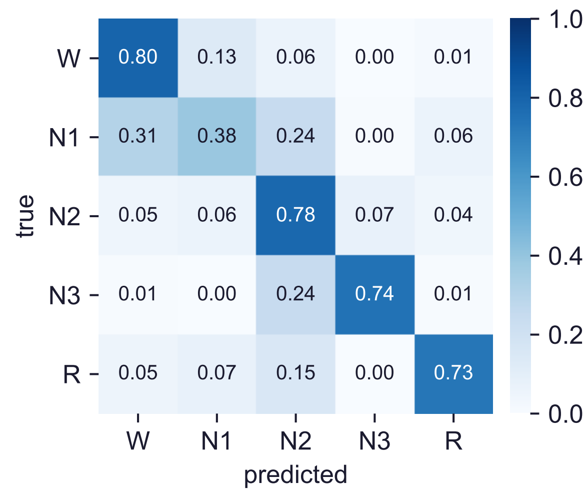

Fig. 4. Confusion matrix (row-normalized). Most error is N1 mislabelled as Wake, N2 or REM, the known ambiguity of transitional sleep.

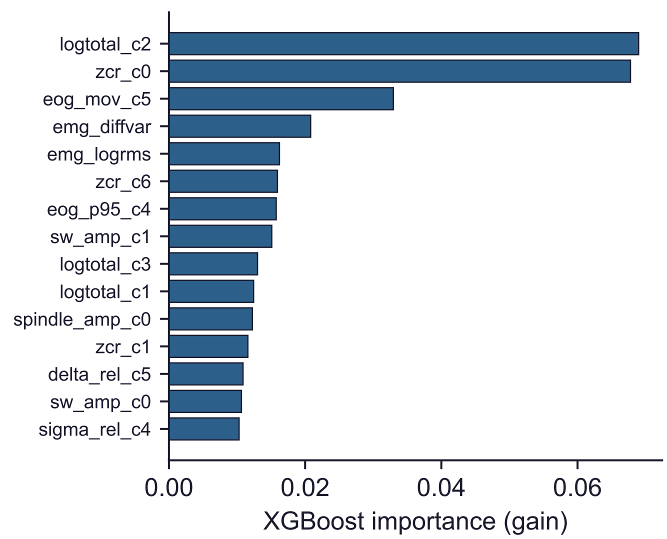

Fig. 5. Top feature importances. Spindle (sigma) power, delta power, and EOG/EMG descriptors dominate, matching AASM scoring physiology.

scattered hard class, bleeding into Wake, N2 and REM. Event features help most on the minority stages because spindle density sharpens N2, slow-wave amplitude sharpens N3, and the EOG/EMG features separate REM from Wake. Because HAG-Net is feature-based, its decisions are auditable: Fig. 5 ranks the most influential features. The ranking is led by broadband power and zero-crossing rate, a proxy for dominant frequency, followed by ocular movement (EOG), chin EMG tone, and slow-wave and spindle amplitude. These are the quantities AASM scoring itself relies on: EOG and EMG separate REM from Wake, slow-wave amplitude marks N3, and spindle activity marks N2. The model therefore decides on clinically meaningful signal rather than on incidental correlates.

E. Calibrated uncertainty

Table VI reports the conformal layer. Empirical coverage matches the target almost exactly (0.900 at α=0.1, 0.950 at α=0.05) with compact sets (average size 1.62 and 2.08; singleton rate 0.51 at α=0.1) and no empty sets, and the base model is already well calibrated (ECE 0.038). Clinically, the prediction set flags the ambiguous N1/REM/Wake epochs for human review instead of forcing an over-confident single label. Black-box stagers do not offer this.

F. Healthy$\to$stroke domain gap

That the stroke setting is genuinely harder, not merely under-modelled, is corroborated concurrently on iSLEEPS by the dataset authors, who report a large generalisation gap and saliency evidence that deep models attend to non-physiological regions in patient data [16]. We quantify it per stage from a fully interpretable angle: a booster trained on healthy Sleep-EDF [41], [42] and evaluated zero-shot on iSLEEPS drops from 0.855 to 0.585 accuracy (-0.27; -0.36 macro-F1; Fig. 6). The collapse is stage-specific: REM recall falls $0.90 \to 0.14$ and N3 $0.84 \to 0.39$, while N2 stays robust ($0.90 \to 0.86$). This mirrors the clinical literature on disrupted REM and slow-wave sleep after stroke. Part of the REM drop is confounded by the frontal-vs-central montage difference between datasets, which we note as a limitation.

G. Lesion-asymmetry biomarker

Because HAG-Net's asymmetry stream is physiological, it supports a clinical readout. For patients with known lesion side we compare sigma-band (spindle) power in the ipsilesional versus contralesional hemisphere over N2 epochs. We observe a directional ipsilesional spindle reduction reaching a trend

TABLE VI
SPLIT-CONFORMAL PREDICTION (LAC). COVERAGE MATCHES TARGET; ECE = 0.038.

α	Target cov.	Empirical cov.	Avg. set size
0.10	0.900	0.900	1.62
0.05	0.950	0.950	2.08

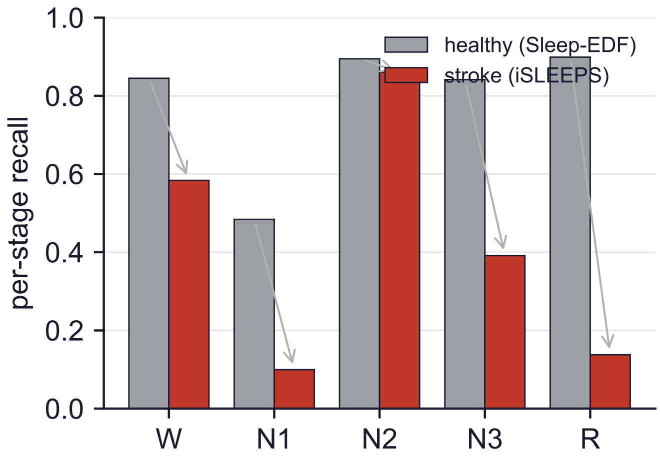

Fig. 6. Zero-shot healthy→stroke transfer. Accuracy falls −0.27, concentrated in REM and N3; N2 is comparatively robust.

($p{\approx}0.065$, $n{=}43$) and, more strikingly, the inter-hemispheric spindle-power asymmetry scales with stroke severity: the asymmetry index correlates with the NIHSS score (Spearman $\rho = 0.41$, $p = 0.006$; Fig. 1(b, c)). This is consistent with a thalamocortical account of spindle generation [13], with asymmetry-based qEEG prognosis [33], and with recent cohort evidence linking spindles to post-stroke cognition and recovery [14], [31], [32]. It shows how an interpretable stager doubles as a lesion-sensitive biomarker. We report this as hypothesis-generating: apnea burden (high in this cohort) and montage are uncontrolled confounds.

H. Per-subject evaluation

Pooled epoch metrics can hide a model that does well on a few long, easy nights. Fig. 7 and Table VII therefore score every patient separately, across the settings we trained. HAG-Net has both the highest median per-patient accuracy (0.755) and the smallest spread (SD 0.091), so its advantage holds patient by patient rather than resting on a favourable subset. The ordering of the four settings matches Table III. Patients at the bottom of the range are hard for every model, which points to recording quality and missing stages rather than to any one architecture. Per-patient macro-F1 is lower than the pooled value for all models because several patients contain no N3 or no REM at all, which caps the per-patient macro average.

I. Statistical significance

Subject-paired tests over the 99 patients confirm the benchmark is not fold-noise. HAG-Net's classical-prior operating point beats the feature-sequence BiLSTM (accuracy $\Delta =$ 0.061, Wilcoxon $p = 8.1 \times 10^{-10}$, Cohen's $d = 0.67$; macro-F1 $p = 6.3 \times 10^{-5}$) and the pure graph/SSM deep variant (accuracy $\Delta = 0.041$, $p = 1.5 \times 10^{-11}$, $d = 0.81$; macro-F1 $p = 1.2{\times}10^{-5}$), all significant after Bonferroni correction. The interpretable-beats-deep result is therefore statistically robust, not anecdotal.

TABLE VII
PER-SUBJECT EVALUATION OVER ALL 99 PATIENTS (OUT-OF-FOLD, WITHOUT HMM): MEAN ± SD, MEDIAN AND RANGE OF PER-PATIENT ACCURACY, AND MEAN PER-PATIENT MACRO-F1.

Setting	Acc (mean±SD)	Med.	Min	Max	mF1
HAG-Net (prior)	**0.737**±0.091	**0.755**	0.347	**0.879**	**0.636**
Graph/SSM deep	0.695±0.098	0.717	0.281	0.841	0.615
Feature-seq BiLSTM	0.676±0.113	0.682	0.361	0.865	0.596
Transfer (Sleep-EDF)	0.604±0.107	0.612	0.305	0.790	0.543

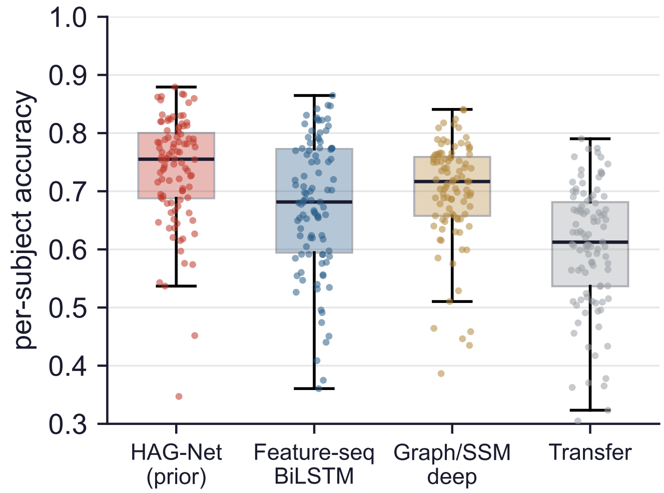

Fig. 7. Per-subject accuracy for each setting, one point per patient; boxes show the median and quartiles. HAG-Net is highest in median and tightest in spread.

J. Discussion

Before drawing conclusions we restate a caveat on the headline comparison. Our results exclude the duplicated recording SN28 and the published baselines do not, so Table III compares $N = 99$ against $N = 100$. We retain the published rows because they are the reference point for this corpus and omitting them would obscure the only external benchmark available, but the reader should treat the accuracy difference of 0.0006 as uninformative: it is a quarter of the estimated duplicate bias and a twenty-fifth of the fold standard deviation. We claim a tie on accuracy, not a win, and we would claim a tie even if the arithmetic favoured us, because the protocols differ. The comparison we do consider sound is against the models trained here under an identical protocol, where the classical prior beats every deep alternative with subject-paired significance.

We draw three conclusions. First, on small clinical cohorts the effective inductive bias is physiological structure rather than network depth; HAG-Net's performance comes from encoding what a scorer reads (spindles, slow waves, atonia, laterality) and adding deep streams only through a residual gate that cannot overfit below the prior. Second, headline accuracy is the wrong target on imbalanced stroke hypnograms; HAG-Net's advantage lies in macro-F1, κ, and calibrated uncertainty, the quantities that govern clinical trust. Third, because the model is interpretable, the same asymmetry representation that guides staging also yields a candidate severity biomarker, so the classifier doubles as a clinical instrument.

VII. Limitations

The study is cross-sectional, single-night and single-scorer; lesion localization is coarse (side/territory); apnea (present in most subjects) and the montage difference in the domain-gap experiment are uncontrolled. Absolute accuracy still trails healthy-cohort systems, reflecting genuine pathological difficulty rather than optimization failure. The biomarker is a within-cohort association, not a validated marker. External transfer from Sleep-EDF did not help (0.62 accuracy), which we attribute to the small, frontally-referenced pretraining corpus.

VIII. Future Work

Natural next steps are: a larger, montage-matched pretraining corpus (SHHS, MASS) to test whether in-domain self-supervision closes the gap where cross-dataset transfer did not; automated, event-level spindle/slow-wave detection feeding the biomarker head; a within-subject, apnea-controlled lesion-mapping study to move the asymmetry biomarker from association toward validation; and prospective evaluation of the conformal review-triage mode with clinicians in the loop.

IX. Conclusion

We presented HAG-Net, a hemispheric asymmetry-guided hybrid for sleep staging on the first public stroke-PSG cohort. Against classical, convolutional, recurrent, transfer and graph/state-space models under a strict patient-exclusive protocol, HAG-Net matches the published deep state of the art on accuracy and exceeds it on the minority-sensitive metrics, while every pure-deep alternative underperforms it ($p < 10^{-9}$). Its design combines a classical prior, a homologous-pair asymmetry graph, a selective state-space decoder, and a residual gate anchored at the prior, and it is built for the data-limited clinical regime rather than against it. Because it is interpretable by construction, the same model exposes a stage-specific domain gap, delivers calibrated uncertainty, and yields a lesion-asymmetry biomarker that tracks stroke severity. The broader lesson is that in high-stakes, small-cohort clinical EEG, physiologically grounded inductive bias and honest uncertainty matter more than raw model capacity.

Author Contributions

Following the CRediT taxonomy. **M.W.S.**: conceptualization, methodology, software, formal analysis, writing (original draft). **E.M.C.A.**: validation, investigation, data curation, writing (review and editing). **M.I.A.S.**: software, visualization, validation, writing (review and editing). All authors read and approved the final manuscript.

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The authors thank the iSLEEPS/NIMHANS team for releasing the dataset. We note the duplicated recording described in Section IV in a constructive spirit; it does not detract from the value of the corpus, which is the first of its kind for this population.

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